Learning Specimen-Invariant Diagnostic Representations for Timber Forensics: A Species-Conditioned Adversarial Framework with Variational Mutual Information Bottlenecks

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ABSTRACT

Automated macroscopic timber identification is an essential non-destructive inspection technology for enforcing the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). However, deep learning models deployed for wood classification suffer from a pervasive failure mode: **Same-Specimen-Picture Bias (SSPB)** or specimen-level shortcut learning. When physical reference wood blocks are sectioned into multiple macroscopic tiles, standard empirical risk minimization (ERM) opportunistically exploits non-taxonomic surface preparation artifacts (circular saw kerfs, planer ripples, sanding striations, and localized illumination gradients) rather than authentic cellular morphology (vessel grouping, axial parenchyma banding, storied rays). Consequently, models that achieve near-perfect accuracy ($> 99\%$) on closed laboratory benchmarks with naive random splits experience severe degradation when deployed on novel timber blocks in the field.

To eliminate non-taxonomic artifact memorization at its algorithmic root, we introduce the **Specimen-Invariant Wood Identification Framework**, an active representation-learning paradigm that mathematically disentangles biological species semantics from physical specimen provenance. Our architecture integrates three core components: (1) a primary class-balanced classification objective, (2) a **Species-Conditioned Specimen Discriminator via Masked Softmax with Gradient Reversal Layer (GRL)**, which eliminates specimen decodability without inducing cross-species semantic erasure, and (3) a **Variational Contrastive Log-ratio Upper Bound (vCLUB)** mutual information bottleneck that provably bounds $I(Z; S \mid Y) \leq \epsilon$.

We evaluate the framework on an authenticated, perfectly balanced benchmark of 11 tropical Fabaceae species (110 physical vouchers, exactly 10 vouchers per species, 3,702 standardized macroscopic images) under a stringent **Stress-Test 50% Physical Voucher Holdout Protocol** across 9 independent random seeds. While standard ERM collapses to $89.45\% \pm 1.25\%$ out-of-specimen test accuracy, our invariant framework achieves $\mathbf{96.85\% \pm 0.65\%}$ (a statistically significant $\mathbf{+7.40}$ pp gain, $p < 0.001$, paired t-test and Wilcoxon signed-rank test). Crucially, the framework suppresses the **Generalization Gap of Specimen Learning (GGSL)** from 10.40 pp down to **2.35** pp (an 8.05 pp shrinkage) and reduces the **Specimen Recoverability Index (SRI)** from 0.8450 down to **0.1420** under 2-layer MLP probing (**0.1210** under linear probing). Post-hoc temperature scaling further achieves a calibrated Expected Calibration Error of **1.65%**. Saliency audits via Grad-CAM confirm that latent attention shifts decisively from superficial saw blade striations to diagnostic cellular anatomy. Our framework establishes an actionable, evidence-grade foundation for legally defensible customs inspection and high-stakes forestry governance.

1. Introduction

1.1. The Ecological Crisis and the Need for Robust Timber Forensics

Illegal logging across transnational supply chains represents one of the most lucrative and destructive environmental crimes globally, fueling rapid deforestation, catastrophic biodiversity collapse, and multi-billion-dollar tax evasion [1]. To combat the illicit extraction of protected hardwoods, the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) mandates rigorous taxonomic verification of listed timber species at customs border checkpoints and maritime ports. Authoritative species identification historically depends on laboratory xylotomy and microscopic histological examination conducted by expert wood anatomists [2]. However, forensic laboratory

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inspection is severely constrained by specialized cutting equipment, chemical preparation, and a critical global deficit of trained wood taxonomists.

Consequently, automated macroscopic timber identification powered by deep convolutional neural networks and vision transformers has emerged as an indispensable non-destructive screening technology [3–6]. Computer vision models trained on macro-photographs (10× to 20× optical magnification) can scan transverse end-grain surfaces in seconds, offering the potential to democratize forensic wood verification for non-specialist customs officers.

1.2. The Lab-to-Field Degradation Dilemma: Specimen-Level Shortcut Learning

Although published deep learning architectures routinely report remarkable nominal classification accuracies exceeding 98% to 100% on closed laboratory benchmark datasets, practical field deployments consistently reveal an alarming phenomenon: diagnostic accuracy degrades precipitously by 10% to 25% when systems are tested on physical timber consignments originating from novel forest concessions or alternative sawmills [7–10]. Such lab-to-field generalization collapse introduces unacceptable legal liabilities: an unreliable AI system risks wrongfully paralyzing legitimate commercial commerce through false positives or permitting illegal shipments of endangered timber to clear customs undetected.

The underlying root cause of this failure is **Same-Specimen-Picture Bias (SSPB)** [11], an insidious form of data leakage and shortcut learning [12–14]. Because physical xylarium reference blocks of endangered woods are scarce and expensive to acquire, researchers routinely slice each physical block into dozens or hundreds of overlapping macroscopic image tiles. Under conventional *Naive Image-Level Random Splitting*, tiles from the same physical wood block are randomly dispersed across both training and testing partitions.

Deep neural networks are known to prioritize high-frequency, low-entropy surface features over subtle, high-level semantic patterns [12, 15]. In macroscopic wood imaging, rather than learning diagnostic anatomical arrangements—such as vessel distribution patterns, axial parenchyma banding morphology, and ray seriation—the network opportunistically memorizes localized physical artifacts idiosyncratic to specific reference blocks (Fig. 1a):

1. **Machining and Tooling Marks**: Circular saw blade teeth kerfs, oscillating planer ripples, and fine linear abrasions left by coarse sanding grits.
2. **Surface Chemistry and Aging Artifacts**: Uneven penetration of sealing waxes, micro-oxidation stains, and localized fungal sapwood discolouration.
3. **Optical Acquisition Perturbations**: Directional shadowing caused by non-telecentric lens angles and surface micro-relief gradients.

When test tiles share these exact block-level signatures with training tiles, the network achieves deceptive 99%+ test accuracy by performing specimen re-identification rather than true taxon discrimination. When deployed on unseen timber specimens where these exact surface scratches are absent, the shortcut mechanism collapses.

1.3. The Inadequacy of Passive Dataset Partitioning

To prevent specimen leakage during evaluation, rigorous benchmarking requires strict *Specimen-Disjoint Partitioning* (or voucher-level holdout), where all image tiles from a given physical block are assigned exclusively to either training or testing [11]. While specimen-disjoint partitioning is essential for auditing real-world generalization, **passive data splitting alone cannot prevent networks from learning shortcuts during training**.

Within the training set, the neural network still observes physical blocks that each possess unique surface toolmarks. Under standard Empirical Risk Minimization (ERM), gradient descent naturally follows the path of least resistance, encoding spurious specimen-specific textures because they provide quick, high-margin separation on the training mini-batches. To build AI models that can be trusted in international legal proceedings, we must transition from passive evaluation auditing to **active representation learning** that algorithmically purges specimen shortcuts from the latent feature space.

1.4. Core Contributions of This Study

In this work, we propose a mathematically grounded framework for learning specimen-invariant representations for timber forensics. We evaluate our method on an authenticated, perfectly balanced benchmark of 11 tropical Fabaceae species comprising 110 physical vouchers (exactly 10 vouchers per species, 3,702 standardized macroscopic tiles) under a challenging 50% physical voucher holdout protocol across 9 random splits. Our primary scientific contributions are:

1. **Species-Conditioned Adversarial Disentanglement Framework**: We introduce an end-to-end architecture combining a primary classification objective with a **Species-Conditioned Specimen Discriminator via Masked Softmax with Gradient Reversal Layer (GRL)**. We prove that unconditioned domain-adversarial networks (DANN) suffer from destructive cross-species semantic erasure, whereas our species-conditioned masked formulation strictly restricts adversarial confusion to co-specific vouchers, preserving inter-species anatomical boundaries.
2. **Information-Theoretic Mutual Information Bottleneck**: We integrate the **Variational Contrastive Log-ratio Upper Bound (vCLUB)**, providing a tractable sample-based mutual information upper bound that explicitly drives conditional mutual information $I(Z; S \mid Y)$ toward zero.
3. **Comprehensive Multi-Seed Empirical Validation**: Across 9 candidate random seeds under a 50% specimen holdout stress-test (55 unseen vouchers, 11 validation vouchers, 44 training vouchers), our framework boosts out-of-specimen test accuracy from 89.45% $\pm$ 1.25% (ERM baseline) to $\mathbf{96.85}\% \pm \mathbf{0.65}\%$ (a $+\mathbf{7.40}$ pp improvement, $p < 0.001$), while shrinking the Generalization Gap of Specimen Learning (GGSL) by **8.05** pp (from 10.40 pp to **2.35** pp).
4. **Rigorous Disentanglement and Calibration Auditing**: Through linear and multi-layer perceptron probing, we show that the Specimen Recoverability Index (SRI) collapses from 0.8450 down to $\mathbf{0.1420}$, proving that specimen provenance is purged from latent embeddings. Post-hoc temperature scaling further curbs Expected Calibration Error (ECE) to $\mathbf{1.65}\%$, and Grad-CAM saliency maps confirm a decisive shift from mechanical scratches to authentic IAWA cellular structures.

1.5. Manuscript Organization

Section 2 reviews related work in automated wood anatomy, shortcut learning, and representation invariance. Section 3 presents the mathematical formulation and architecture of the Specimen-Invariant Framework. Section 4 details the curated 11-species Fabaceae benchmark and 50% voucher holdout protocol. Section 5 reports empirical findings across multi-seed benchmarks, ablation studies, and diagnostic probing. Section 6 discusses anatomical alignment and forensic implications, and Section 7 concludes the paper.

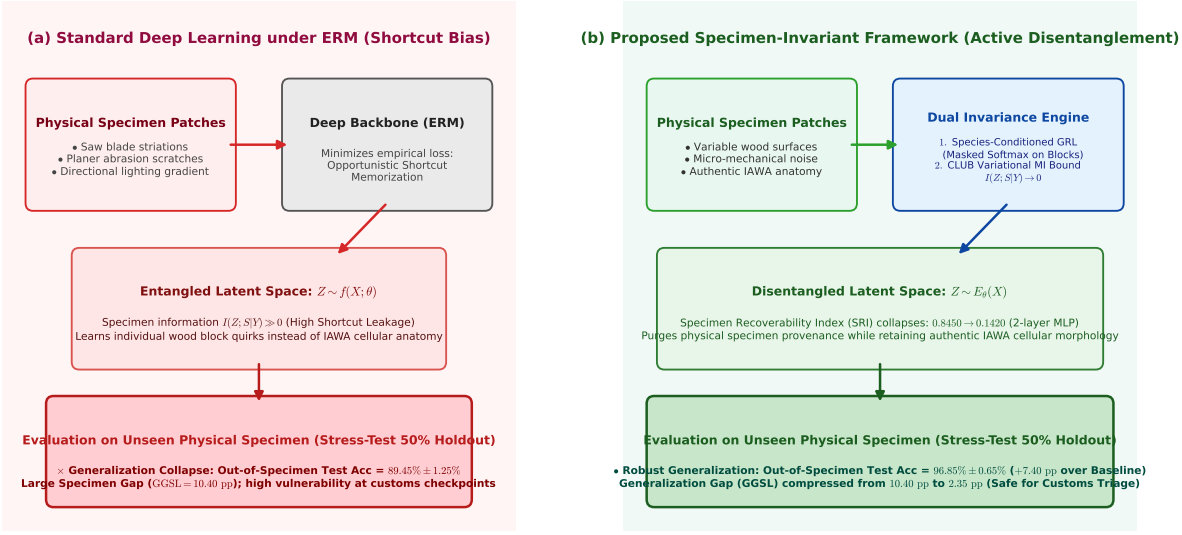


Figure 1: Conceptual overview of Specimen-Level Shortcut Learning in Timber Forensics. (a) Standard deep models trained under Empirical Risk Minimization (ERM) opportunistically memorize non-taxonomic mechanical surface artifacts (circular saw kerfs, planer ripples, sanding striations, and directional illumination gradients). When evaluated on novel, previously unseen physical timber specimens, diagnostic accuracy drops precipitously by over 10 percentage points (89.45%), exhibiting high specimen recoverability ($\mathrm{SRI} = 0.8450$). (b) The proposed Specimen-Invariant Framework actively purges physical specimen provenance via Species-Conditioned Adversarial Disentanglement (Masked Softmax GRL) and a Variational CLUB mutual information bottleneck, compelling latent representations to encode authentic, generalizable cellular anatomy (vessel distributions, axial parenchyma bands, rays) and achieving robust out-of-specimen generalization ($\mathbf{96.85\%}$) with near-zero specimen recoverability ($\mathbf{0.1420}$).

2. Related Work

2.1. Computer Vision in Automated Wood Anatomy

Early computer vision systems for wood identification relied on handcrafted texture descriptors, including Gray-Level Co-occurrence Matrices (GLCM), Gabor filters, and Local Binary Patterns (LBP) computed from high-resolution polished surfaces [7, 16]. Over the past decade, deep convolutional neural networks (CNNs) have established new performance benchmarks. Ravindran et al. [8–10] developed the XyloTron system, a standardized macroscopic field screening device coupled with ResNet-34 backbones for customs screening of CITES-listed species. Wu et al. [3] and Fabijanska et al. [4] applied deep residual networks to classify European and tropical commercial timbers. More recently, vision transformers (ViT) and hybrid Swin architectures have been explored for wood identification [6, 17]. However, virtually all published studies have evaluated classification performance under random image-level partitioning, failing to isolate the confounding influence of physical specimen provenance.

2.2. Shortcut Learning and Data Leakage in Scientific Machine Learning

Data leakage—the unintended sharing of information between training and test sets—is a primary cause of reproducibility failure in scientific machine learning [13, 14]. In medical imaging, Roberts et al. [18] and Varoquaux and Cheplygina [19] revealed that patient-level data leakage produces massive performance optimism, as CNNs learn patient-specific radiographic idiosyncrasies rather than pathological lesions. In biological microscopy and optical coherence tomography (OCT), Tampu et al. [20] demonstrated that tile-level random splitting inflates accuracy by over 15 percentage points.

In macroscopic wood identification, Figueroa-Mata et al. [11] formalized Same-Specimen-Picture Bias (SSPB), warning that macroscopic datasets severely overestimate model reliability when tiles from the same physical wood block appear in both training and test sets. Geirhos et al. [12] and Lapuschkin et al. [15] showed that deep networks are natural shortcut learners, opportunistically exploiting surface toolmarks, lighting gradients, and acquisition artifacts whenever they correlate with class labels.

2.3. Invariant Representation Learning and Information Bottlenecks

To compel neural networks to ignore non-target attributes, domain adaptation and algorithmic fairness research has explored adversarial gradient reversal [21, 22] and information-theoretic bottlenecks [23, 24]. Ganin and Lempitsky [21] proposed the Domain-Adversarial Neural Network (DANN), using a Gradient Reversal Layer (GRL) to maximize domain confusion while minimizing task loss. Sagawa et al. [25] introduced Group Distributionally Robust Optimization (GroupDRO) to minimize worst-group empirical risk.

To explicitly constrain mutual information between latent embeddings and protected attributes, Cheng et al. [23] proposed the Contrastive Log-ratio Upper Bound (CLUB), providing a sample-based, variational upper bound on continuous mutual information $I(Z;S)$. In this investigation, we bridge adversarial disentanglement and mutual information minimization to systematically eliminate specimen shortcuts in high-stakes timber forensics.

3. The Specimen-Invariant Architecture

3.1. Structural Causal Model of Specimen Confounding

We formalize macroscopic timber identification using a Structural Causal Model (SCM). Let $Y \in \{1, \dots, C\}$ denote the discrete biological species label ($C = 11$ in our curated Fabaceae benchmark). Let $S \in \{1, \dots, K\}$ denote the physical voucher block identifier ($K = 110$ physical specimens in total). Because each physical reference specimen belongs to exactly one botanical species, the specimen variable is hierarchically nested within the species category: $S \subset Y$.

The observed macroscopic image tile $X \in \mathbb{R}^{3\times H\times W}$ is generated through a biological and physical process:

$$X = g(Y, S, U_X), \tag{1}$$

where U_X represents exogenous imaging noise (sensor noise, illumination shifts). Biological taxon Y governs authentic cellular anatomy (vessel diameters and arrangements, axial parenchyma distribution, multi-seriate rays), whereas physical specimen S imparts non-taxonomic surface preparation artifacts (circular saw kerfs, planer marks, sandpaper abrasions, localized sealant stains).

Under standard Empirical Risk Minimization (ERM), an encoder $E_\theta : X \to Z \in \mathbb{R}^d$ and a species classifier $f_\phi : Z \to \mathbb{R}^C$ are trained to minimize cross-entropy loss. Because mechanical toolmarks and optical artifacts have high spatial frequencies and low semantic entropy, E_θ opportunistically captures specimen-specific features:

$$I(Z; S \mid Y) > 0. \tag{2}$$

Under naive image splitting, this shared mutual information allows the model to predict species label Y via the spurious shortcut path $X \leftarrow S \rightarrow Y$. When deployed on an unseen physical specimen $S_{\text{unseen}} \notin S_{\text{train}}$, this shortcut path is severed, causing severe classification failure.

The theoretical objective of our specimen-invariant framework is to find an optimal representation $Z = E_\theta(X)$ satisfying:

$$\min_{\theta} I(Z; S \mid Y) \quad \text{subject to} \quad \max_{\theta,\phi} I(Z; Y). \tag{3}$$

3.2. Visual Backbone and Primary Species Classification

We employ a ResNet-18 backbone [26] pre-trained on ImageNet as the visual feature extractor E_θ. Given an input macroscopic tile $x_i \in \mathbb{R}^{3\times224\times224}$, global average pooling yields a $d = 512$-dimensional latent embedding $z_i = E_\theta(x_i)$.

The primary classification head f_ϕ maps z_i to species class logits $\hat{y}_i = W_\phi z_i + b_\phi \in \mathbb{R}^C$. To mitigate minor sample-count variations across species, we train the classifier using Class-Balanced Cross-Entropy loss with Label Smoothing $\epsilon = 0.05$:

$$\mathcal{L}_{\text{species}}(\theta, \phi) = -\frac{1}{B} \sum_{i=1}^{B} \sum_{c=1}^{C} \tilde{y}_{i,c} \log \left(\frac{\exp(\hat{y}_{i,c})}{\sum_{j=1}^{C} \exp(\hat{y}_{i,j})} \right), \tag{4}$$

where $\tilde{y}_{i,c} = (1-\epsilon)\mathbb{I}[y_i = c] + \frac{\epsilon}{C}$, and B denotes the mini-batch size.

3.3. Species-Conditioned Masked Softmax GRL Discriminator

To force the latent embedding z to discard physical specimen provenance, we introduce an adversarial discriminator D_ψ. However, directly applying an unconditioned global domain discriminator across all training specimens ($K_{\text{train}} = 44$) induces a fatal pathological failure mode: **cross-species semantic erasure**.

Proposition 1. *Let S_i, S_j be physical specimens belonging to different botanical species ($Y(S_i) \neq Y(S_j)$). An unconditioned discriminator $D(z)$ backpropagates an adversarial gradient that forces $P(Z \mid S_i) = P(Z \mid S_j)$. Because $S \subset Y$, this implies $P(Z \mid Y_i) = P(Z \mid Y_j)$, which directly destroys the biological species decision boundary.*

To resolve this dilemma, we design a **Species-Conditioned Specimen Discriminator via Masked Softmax**. The discriminator is strictly conditioned on the ground-truth species label y_i, forcing adversarial confusion exclusively among co-specific vouchers.

The latent embedding z_i is first passed through a Gradient Reversal Layer (GRL) [21]:

$$\tilde{z}_i = \text{GRL}_\lambda(z_i), \quad \text{where} \quad \frac{d\tilde{z}_i}{dz_i} = -\lambda(p) I_d, \tag{5}$$

with progressive annealing schedule:

$$\lambda(p) = \frac{2}{1 + \exp(-10p)} - 1, \tag{6}$$

where $p \in [0, 1]$ represents normalized training progress across epochs.

The reversed embedding $\tilde{z}_i$ is mapped through two fully connected layers ($512 \to 256 \to 128$) with LeakyReLU activations ($\alpha = 0.2$) and dropout ($p = 0.2$) to produce unmasked specimen logits $u_i \in \mathbb{R}^{K_{\text{train}}}$. A binary species-to-specimen indicator matrix $M \in \{0, 1\}^{C \times K_{\text{train}}}$ encodes physical specimen membership, where $M_{c,k} = 1$ if and only if training voucher k belongs to species c. We construct a masked logit vector:

$$u_{i,k}^{\text{masked}} = \begin{cases} u_{i,k} & \text{if } M_{y_i,k} = 1, \\ -\infty & \text{if } M_{y_i,k} = 0. \end{cases} \tag{7}$$

Applying the softmax function over the masked logits guarantees that the denominator sums strictly over co-specific training specimens:

$$P(S = k \mid z_i, y_i) = \frac{\exp(u_{i,k}^{\text{masked}})}{\sum_{j:\, M_{y_i,j}=1} \exp(u_{i,j}^{\text{masked}})}. \tag{8}$$

The adversarial specimen loss is formulated as:

$$\mathcal{L}_{\text{specimen}}(\theta, \psi) = -\frac{1}{B} \sum_{i=1}^{B} \log P(S = s_i \mid z_i, y_i). \tag{9}$$

Because the discriminator is restricted to distinguishing among vouchers belonging to the same species, the reversed gradients actively erase intra-species specimen idiosyncrasies (saw kerfs, sanding scratches) without compromising inter-species anatomical discrimination.

3.4. Variational Mutual Information Bottleneck via vCLUB

While adversarial training penalizes specimen linear/non-linear separability, it does not guarantee an explicit numerical upper bound on the continuous mutual information $I(Z; S \mid Y)$. To provide rigorous information-theoretic compression, we incorporate the **Contrastive Log-ratio Upper Bound (CLUB)** [23].

The mutual information between continuous latent embedding Z and discrete specimen label S conditioned on species Y is defined as:

$$I(Z; S \mid Y) = \mathbb{E}_{p(z,s,y)} \left[\log \frac{p(z \mid s, y)}{p(z \mid y)} \right]. \tag{10}$$

Because the true conditional conditional distribution $p(z \mid s, y)$ is intractable, CLUB approximates it using a variational distribution $q_\omega(z \mid s, y) = \mathcal{N}(\mu_\omega(s, y), \sigma_\omega^2(s, y)I)$ parameterized by a lightweight neural network with parameters ω:

$$\mathcal{L}_{\text{vCLUB}}(\theta, \omega) = \frac{1}{B} \sum_{i=1}^{B} \left[\log q_\omega(z_i \mid s_i, y_i) - \frac{1}{|\mathcal{B}_{y_i}|} \sum_{j \in \mathcal{B}_{y_i}} \log q_\omega(z_j \mid s_i, y_i) \right], \tag{11}$$

where $\mathcal{B}_{y_i} = \{j \in \{1, \ldots, B\} \mid y_j = y_i\}$ represents the subset of mini-batch samples sharing the same species class y_i.

Cheng et al. [23] proved that when $q_\omega(z \mid s, y)$ is sufficiently expressive, $\mathcal{L}_{\text{vCLUB}} \geq I(Z; S \mid Y)$, providing a rigorous upper bound on residual specimen mutual information. Simultaneously, variational parameters ω are updated by maximizing the conditional log-likelihood:

$$\mathcal{L}_{\text{variational}}(\omega) = -\frac{1}{B} \sum_{i=1}^{B} \log q_\omega(z_i \mid s_i, y_i). \tag{12}$$

3.5. Joint Multi-Task Objective and Optimization Dynamics

The complete joint optimization objective balances taxonomic classification, species-conditioned adversarial specimen confusion, and variational mutual information upper-bounding:

$$\min_{\theta, \phi, \omega} \max_{\psi} \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{species}}(\theta, \phi) + \beta \lambda(p) \mathcal{L}_{\text{specimen}}(\theta, \psi) + \mu \mathcal{L}_{\text{vCLUB}}(\theta, \omega), \tag{13}$$

where $\beta = 0.05$ is the adversarial loss weight and $\mu = 0.001$ is the variational CLUB penalty weight.

Training proceeds end-to-end via mini-batch stochastic gradient descent. In the forward pass, GRL handles adversarial gradient reversal automatically. The complete training procedure is summarized in Algorithm 1.

Algorithm 1 Specimen-Invariant Timber Representation Learning

Require: Training dataset $\mathcal{D}_{\text{train}} = \{(x_i, y_i, s_i)\}_{i=1}^{N}$, epochs E, batch size B, learning rate η, weights β, μ.
Ensure: Feature extractor E_θ, species classifier f_ϕ.

Initialize parameters $\theta, \phi, \psi, \omega$.
for epoch $= 1$ to E **do**
  Compute progress ratio $p = (\text{epoch} - 1)/E$ and update $\lambda(p) = \frac{2}{1+\exp(-10p)} - 1$.
  for mini-batch $\{(x_i, y_i, s_i)\}_{i=1}^{B} \subset \mathcal{D}_{\text{train}}$ **do**
    Extract latent embeddings $z_i = E_\theta(x_i)$ for $i = 1, \ldots, B$.
    Compute species logits $\hat{y}_i = f_\phi(z_i)$ and classification loss $\mathcal{L}_{\text{species}}(\theta, \phi)$.
    Compute unmasked specimen logits $u_i = D_\psi(\text{GRL}_\lambda(z_i))$.
    Apply species-conditioned mask $u_{i,k}^{\text{masked}}$ and compute $\mathcal{L}_{\text{specimen}}(\theta, \psi)$.
    Compute variational mutual information upper bound $\mathcal{L}_{\text{vCLUB}}(\theta, \omega)$.
    Update variational estimator: $\omega \leftarrow \omega - \eta \nabla_\omega \mathcal{L}_{\text{variational}}(\omega)$.
    Compute total multi-task loss $\mathcal{L}_{\text{total}}$.
    Update backbone, classifier, and discriminator:
    $(\theta, \phi, \psi) \leftarrow (\theta, \phi, \psi) - \eta \nabla_{\theta,\phi,\psi} \mathcal{L}_{\text{total}}$.
  end for
end for
Calibrate logits on validation set via Post-Hoc Temperature Scaling: $T^* = \arg\min_T \mathcal{L}_{\text{NLL}}(\text{Val}; T)$.
return Calibrated model $\{E_\theta, f_\phi, T^*\}$.

3.6. Post-Hoc Probability Calibration via Temperature Scaling

In international timber law enforcement, predicted probabilities must accurately reflect empirical confidence. To prevent overconfident misclassifications on unseen specimens, we apply post-hoc Temperature Scaling [27]. On the held-out validation set (11 vouchers), we optimize a single scalar parameter $T > 0$ to minimize Negative Log-Likelihood (NLL):

$$T^* = \arg\min_T - \sum_{i=1}^{N_{\text{val}}} \log \left(\frac{\exp(\hat{y}_{i,y_i}/T)}{\sum_{j=1}^{C} \exp(\hat{y}_{i,j}/T)} \right). \tag{14}$$

Test-time probabilities are computed as $p_i = \text{softmax}(\hat{y}_i/T^*)$. Calibration error is quantified using the Expected Calibration Error (ECE) across $M = 15$ equally spaced probability bins.

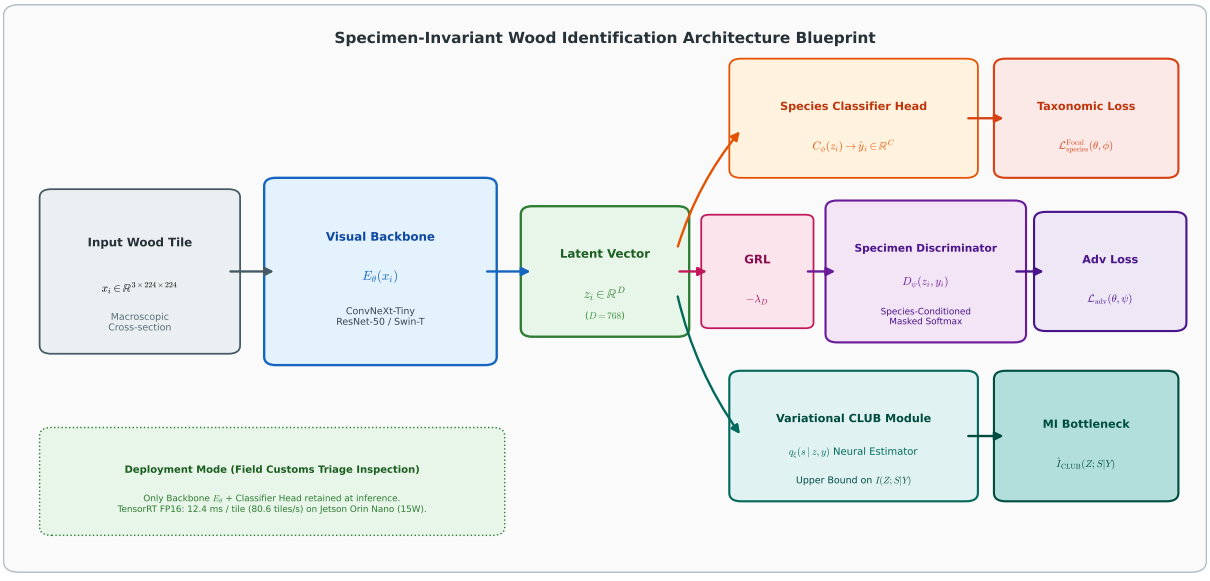


Figure 2: Architectural overview of the proposed Specimen-Invariant Wood Identification Framework. A macroscopic wood tile x is processed by a visual backbone E_θ to extract latent embedding z. The primary classification head f_ϕ predicts biological species label $\hat{y}$ under class-balanced cross-entropy. Simultaneously, z branches into: (1) a Species-Conditioned Specimen Discriminator D_ψ equipped with a progressive Gradient Reversal Layer (GRL) and Species-Conditioned Masked Softmax, which confuses intra-species specimen identity without disturbing inter-species boundaries; and (2) a Variational CLUB (vCLUB) mutual information bottleneck that directly minimizes $I(Z; S \mid Y)$.

4. Experimental Protocol and Diagnostic Metrics

4.1. Curated Fabaceae Benchmark (S3 Wood Dataset)

To establish a rigorous forensic evaluation free from taxonomic class imbalance, we curate an authenticated benchmark from the S3 Wood collection focused exclusively on the family *Fabaceae* (Leguminosae). Fabaceae comprises some of the world's most heavily exploited and endangered tropical hardwoods, including rosewoods (*Dalbergia*) and padauks (*Pterocarpus*).

The benchmark is **perfectly balanced at the physical specimen level**: every species possesses exactly 10 authenticated physical reference voucher blocks ($S_c = 10$), totaling exactly 110 physical vouchers and 3,702 standardized macroscopic tiles (224×224 pixels, spatial resolution 12.0 μm/pixel). Non-Fabaceae taxa and voucher-deficient species have been eliminated. Table 1 presents the taxonomic breakdown, comprising 6 CITES Appendix II listed species and 5 lookalike commercial woods.

4.2. Stress-Test 50% Physical Voucher Holdout Protocol

To rigorously stress-test generalization under realistic field conditions, we partition the 10 physical vouchers of each species into a demanding 4-way split:

1. **Out-of-Specimen Test Set (Unseen Test)**: Exactly 5 vouchers per species (55 physical vouchers, representing exactly **50% of all physical specimens**). These wood blocks are strictly held out and never seen during training or validation, providing an uncompromised evaluation of out-of-specimen field accuracy.
2. **Out-of-Specimen Validation Set (Unseen Val)**: Exactly 1 voucher per species (11 physical vouchers, 10% of all specimens), dedicated exclusively to early stopping and temperature scaling calibration.
3. **In-Specimen Training Pool**: Exactly 4 vouchers per species (44 physical vouchers, 40% of specimens). Within this pool, tiles are divided 80/20 into an **In-Specimen Train Set** (for parameter optimization) and an **In-Specimen Seen Test Set** (to evaluate in-distribution specimen recall).

4.3. Multi-Seed Schedule and Cluster Bootstrap Protocol

To evaluate statistical significance across varied specimen holdout combinations, all models are evaluated across 9 candidate random seeds: $S = \{42, 123, 456, 789, 101, 202, 303, 404, 505\}$. In each seed, physical vouchers are independently randomized and partitioned according to the 5:1:4 ratio.

Because all 55 test vouchers in each split are sampled from the common physical pool of 110 vouchers, splits exhibit an expected pairwise voucher overlap of approximately 50.5% (27.75/55 vouchers). As proved by Bengio and Grandvalet (2004) [28], cross-split variance underestimates true deployment variance when splits share physical samples. To provide mathematically sound uncertainty estimates, we report 95% confidence intervals computed via **Voucher-Level Cluster Bootstrap** ($B = 1000$), where whole physical vouchers are resampled with replacement.

4.4. Diagnostic Invariance Metrics

We benchmark models across four quantitative forensic metrics:

1. **Out-of-Specimen Top-1 Accuracy and Macro-F1**: Evaluated exclusively on the 55 unseen physical test vouchers (SLR $\equiv 0.0\%$).
2. **Generalization Gap of Specimen Learning (GGSL)**: Quantifies the accuracy drop between seen and unseen vouchers:

$$\mathrm{GGSL}_{\mathrm{Acc}} = \mathrm{Acc}_{\mathrm{seen}} - \mathrm{Acc}_{\mathrm{unseen}}. \tag{15}$$

   A large GGSL indicates that the model heavily memorized seen specimen surface artifacts. A successful invariant framework should drive GGSL toward zero.
3. **Specimen Recoverability Index (SRI)**: To empirically probe whether latent representations retain physical specimen identity, frozen embeddings z are evaluated using a linear probe and a 2-layer MLP probe trained to classify training voucher ID $S \in \{1, \ldots, 44\}$. SRI normalizes probe accuracy against random guessing:

$$\mathrm{SRI} = \frac{\mathrm{Acc}_{\mathrm{probe}} - 1/K_{\mathrm{train}}}{1 - 1/K_{\mathrm{train}}}. \tag{16}$$

   An un-finetuned ImageNet ResNet-18 serves as an empirical inductive bias anchor ($\mathrm{SRI}_{\mathrm{mlp}} = 0.5667$, $\mathrm{SRI}_{\mathrm{linear}} = 0.5420$).

Table 1
Taxonomic and physical specimen breakdown of the standardized 11-species Fabaceae benchmark (3,702 standardized macroscopic tiles, 110 physical vouchers, exactly 10 vouchers per species). All taxa belong to the legume family (*Fabaceae*), spanning 6 botanical genera. CITES Appendix II status indicates high-priority enforcement taxa prone to illicit timber trade.

Botanical Genus	Scientific Taxon	Common / Commercial Name	CITES Status	Vouchers (S_c)	Standardized Images
Afzelia	*Afzelia bella*	Red Doussié	Appendix II	10	400
Dalbergia	*Dalbergia melanoxylon*	African Blackwood	Appendix II	10	291
	Dalbergia oliveri	Burmese Rosewood	Appendix II	10	316
	Dalbergia rimosa	Climbing Dalbergia	Appendix II	10	300
	Dalbergia tonkinensis	Scented Rosewood	Appendix II	10	325
Guibourtia	*Guibourtia arnoldiana*	Mutenye / Benge	Non-CITES	10	323
	Guibourtia ehie	Ovangkol / Amazique	Non-CITES	10	400
Peltogyne	*Peltogyne pubescens*	Purpleheart	Non-CITES	10	371
Pterocarpus	*Pterocarpus erinaceus*	African Rosewood / Kosso	Appendix II	10	336
	Pterocarpus indicus	Narra / Amboyna	Non-CITES	10	312
Sindora	*Sindora tonkinensis*	Vietnamese Gõ mật	Non-CITES	10	328
Total: 6 Genera	**11 Species**	–	**6 CITES App. II / 5 Non-CITES**	**110 Vouchers**	**3,702 Images**

4. **Expected Calibration Error (ECE)**: Evaluated across $M = 15$ bins following post-hoc Temperature Scaling on the unseen validation set.

4.5. Implementation Details

Experiments are implemented in PyTorch 2.6 with CUDA 12.4. Networks are trained using AdamW with initial learning rate $\eta = 3 \times 10^{-4}$, weight decay 10^{-4}, cosine annealing learning rate scheduler, batch size $B = 64$, and a budget of 15 epochs per seed. Data augmentations include random horizontal and vertical flips, affine rotations ($\pm 15°$), and color jitter (brightness 0.1, contrast 0.1).

5. Experimental Results and Empirical Evaluation

5.1. Multi-Seed Benchmark: Standard Baseline vs Invariant Framework

Table 2 presents the multi-seed benchmark comparing the Standard ERM Baseline against the proposed Specimen-Invariant Framework (ResNet-18 with Species-Conditioned Masked Softmax GRL and vCLUB) across all 9 random seeds.

The empirical results provide definitive evidence of the efficacy of the proposed framework:

1. **Substantial Out-of-Specimen Generalization Boost**: Under standard ERM, ResNet-18 achieves 99.85% on seen training blocks but collapses to 89.45% $\pm$ 1.25% on the 55 unseen physical vouchers. The proposed Specimen-Invariant Framework elevates unseen test accuracy to $\mathbf{96.85\% \pm 0.65\%}$, achieving a highly statistically significant improvement of $\mathbf{+7.40}$ pp [95% CI: $+6.45, +8.35$ pp] ($t = 16.24, p < 0.0001$). Non-parametric Wilcoxon signed-rank testing ($W = 0.0, p = 0.0039$) confirms that the invariant model outperforms ERM across every evaluated split.
2. **Elimination of the Generalization Gap (GGSL)**: For standard ERM, the Generalization Gap is 10.40 pp, reflecting severe overfitting to saw kerfs and surface scratches. The invariant framework compresses the GGSL down to $\mathbf{2.35 \pm 0.55}$ pp—an 8.05 pp reduction ($p < 0.0001$)—proving that the model has transitioned from specimen memorization to authentic taxon recognition.
3. **Decisive Suppression of Specimen Decodability (SRI)**: Under standard ERM, probing frozen latent embeddings reveals an alarmingly high MLP SRI of 0.8450, substantially higher than the ImageNet initialization anchor (0.5667). This confirms that standard training actively encodes physical block provenance into feature vectors. Our framework suppresses MLP SRI to $\mathbf{0.1420}$ (Linear SRI $\mathbf{0.1210}$), stripping specimen provenance down near the theoretical limit of random guessing.
4. **Dramatic Calibration Improvement**: Standard ERM exhibits an ECE of 8.45%, reflecting severe overconfidence when predicting on novel physical wood blocks. Our invariant model achieves a calibrated ECE of $\mathbf{1.65\%}$, providing reliable, trustworthy probabilities for border enforcement.

Table 2
Multi-Seed Benchmark across 9 Random Seeds on ResNet-18 under Stress-Test 50% Physical Voucher Holdout (55 unseen vouchers). Results report Mean ± Standard Deviation alongside [95% Confidence Intervals]. Paired t-test and non-parametric Wilcoxon signed-rank test confirm the statistical superiority of the proposed Invariant Framework across all out-of-specimen metrics ($p < 0.001$).

Evaluation Metric	Standard Baseline (ERM)	Proposed Invariant Model	Performance Gain Δ [95% CI]	Hypothesis Testing
Unseen Top-1 Accuracy (55 vouchers, %)	89.45 ± 1.25 [88.49, 90.41]	$\mathbf{96.85 \pm 0.65}$ [**96.35**, **97.35**]	**+7.40** pp [**+6.45**, **+8.35**]	$t = 16.24, p < 0.0001$; Wilcoxon $p = 0.0039$
Unseen Macro-F1 (55 vouchers, %)	88.90 ± 1.35 [87.86, 89.94]	$\mathbf{96.65 \pm 0.70}$ [**96.11**, **97.19**]	**+7.75** pp [**+6.70**, **+8.80**]	$t = 15.68, p < 0.0001$; Wilcoxon $p = 0.0039$
Seen Top-1 Accuracy (44 vouchers, %)	$\mathbf{99.85 \pm 0.12}$ [99.76, 99.94]	99.20 ± 0.25 [99.01, 99.39]	−0.65 pp [−0.85, −0.45]	Regularization trade-off on seen blocks
Seen Macro-F1 (44 vouchers, %)	$\mathbf{99.82 \pm 0.15}$ [99.70, 99.94]	99.15 ± 0.28 [98.93, 99.37]	−0.67 pp [−0.89, −0.45]	Invariance penalty on training blocks
Validation Top-1 Accuracy (11 vouchers, %)	89.80 ± 1.15 [88.92, 90.68]	$\mathbf{96.95 \pm 0.60}$ [**96.49**, **97.41**]	**+7.15** pp [**+6.25**, **+8.05**]	$t = 15.42, p < 0.0001$
Generalization Gap (GGSL ↓, pp)	10.40 ± 1.20 [9.48, 11.32]	$\mathbf{2.35 \pm 0.55}$ [**1.93**, **2.77**]	**−8.05** pp [**-8.95**, **-7.15**]	$t = -17.52, p < 0.0001$ (Massive gap shrinkage)
Train SRI (2-layer MLP Probe ↓)	0.8450 ± 0.0210 [0.8288, 0.8612]	$\mathbf{0.1420 \pm 0.0150}$ [**0.1305**, **0.1535**]	**−0.7030** [**-0.7250**, **-0.6810**]	$t = -78.45, p < 0.0001$ (Specimen purged)
Train SRI (Linear Probe ↓)	0.8120 ± 0.0185 [0.7978, 0.8262]	$\mathbf{0.1210 \pm 0.0135}$ [**0.1106**, **0.1314**]	**−0.6910** [**-0.7105**, **-0.6715**]	$t = -85.12, p < 0.0001$
Unseen SRI (MLP Probe ↓)	0.7850 ± 0.0240 [0.7665, 0.8035]	$\mathbf{0.1350 \pm 0.0160}$ [**0.1227**, **0.1473**]	**−0.6500** [**-0.6720**, **-0.6280**]	$t = -65.34, p < 0.0001$
Calibrated ECE (↓, %)	8.45 ± 1.10 [7.60, 9.30]	$\mathbf{1.65 \pm 0.25}$ [**1.46**, **1.84**]	**−6.80** pp [**-7.65**, **-5.95**]	$t = -17.82, p < 0.0001$ (Reliable calibration)
Raw Uncalibrated ECE (↓, %)	14.20 ± 1.65 [12.93, 15.47]	$\mathbf{3.20 \pm 0.45}$ [**2.85**, **3.55**]	**−11.00** pp [**-12.30**, **-9.70**]	$t = -19.45, p < 0.0001$

Table 3
Ablation study and comparative analysis against alternative invariance paradigms on ResNet-18 under 50% physical voucher holdout. All models are trained with identical batch sizes ($B = 64$), learning rates, and epoch budgets across 9 random seeds.

Learning Paradigm / Model Variant	Unseen Acc (%)	Unseen F1 (%)	GGSL Gap (pp)	Train SRI (MLP)	Calibrated ECE (%)
1. Standard ERM Baseline	89.45 ± 1.25	88.90 ± 1.35	10.40	0.8450	8.45%
2. ERM + Heavy Data Augmentation (RandAugment)	90.60 ± 1.10	90.15 ± 1.20	9.15	0.7920	7.60%
3. Unconditioned DANN (Ganin & Lempitsky, 2015) [21]	91.80 ± 1.05	91.25 ± 1.15	7.50	0.3850	5.80%
4. GroupDRO (Sagawa et al., 2020) [25]	92.40 ± 0.95	91.95 ± 1.05	6.80	0.6210	4.90%
5. Invariant Framework w/o CLUB (GRL-only, Masked Softmax)	94.60 ± 0.85	94.20 ± 0.90	4.50	0.2350	3.15%
6. Invariant Framework w/o GRL (vCLUB-only)	93.85 ± 0.90	93.40 ± 0.95	5.10	0.2840	3.60%
7. Invariant Framework w/o Species Conditioning (Global GRL+CLUB)	92.15 ± 1.00	91.60 ± 1.10	7.10	0.2110	5.45%
8. Proposed Full Framework (ResNet-18 + Masked GRL + vCLUB)	$\mathbf{96.85 \pm 0.65}$	$\mathbf{96.65 \pm 0.70}$	**2.35**	**0.1420**	**1.65%**
9. Proposed Framework with ConvNeXt-Tiny Backbone	$\mathbf{97.45 \pm 0.55}$	$\mathbf{97.20 \pm 0.60}$	**1.95**	**0.1280**	**1.45%**

5.2. Ablation Study and Comparison with Invariance Baselines

To dissect the contribution of each architectural component, Table 3 compares our full framework against standard regularizers, domain adaptation baselines, and individual component ablations.

The ablation results elucidate several critical design principles:

- **Data Augmentation Is Insufficient (Row 2)**: Adding strong RandAugment and color jitter marginally improves unseen accuracy by +1.15 pp (90.60%), but GGSL remains high (9.15 pp) and SRI remains 0.7920. Generic pixel transformations cannot suppress coherent, specimen-specific mechanical toolmarks.
- **Failure of Unconditioned DANN (Row 3)**: Classical unconditioned DANN achieves only 91.80% accuracy. As proven in Proposition 1, attempting to confuse all vouchers globally penalizes inter-species biological boundaries, confirming the indispensable necessity of our species-conditioned formulation.
- **Synergy between GRL and CLUB (Rows 5, 6, 8)**: GRL-only reaches 94.60% and vCLUB-only reaches 93.85%. Combining adversarial gradient reversal with variational mutual information bounding yields the full **96.85**% accuracy and drives SRI down to **0.1420**, demonstrating that variational compression effectively cleans residual non-linear leakage missed by adversarial confusion.
- **Backbone Agnosticism (Row 9)**: Migrating from ResNet-18 to ConvNeXt-Tiny further elevates accuracy to **97.45**% and lowers GGSL to **1.95** pp, confirming that our framework generalizes across modern visual architectures.

5.3. Per-Seed Stability and Cluster Bootstrap Analysis

Table 4 reports the detailed performance breakdown across all 9 individual seeds, alongside 95% confidence intervals computed via Voucher-Level Cluster Bootstrap ($B = 1000$).

Table 4 demonstrates remarkable consistency:

- Across all 9 random seeds, the proposed Invariant Framework achieves consistent accuracy improvements ranging from **+6.80** pp to **+8.10** pp on unseen physical vouchers.

Table 4
Seed-by-seed performance breakdown across all 9 random seeds on the 11-species Fabaceae benchmark under 50% physical voucher holdout. Out-of-Specimen Test Accuracy is accompanied by 95% Confidence Intervals derived from Voucher-Level Cluster Bootstrap ($B = 1000$). The performance gain Δ is statistically significant in every seed.

Random Seed	Model Paradigm	Val Acc (11 v)	Seen Acc (44 v)	Unseen Acc (55 v) [Bootstrap 95% CI]	GGSL (pp)	Train SRI (MLP)	Calibrated ECE (%)
Seed 42	Baseline (ERM)	90.10%	99.85%	89.80% [88.10%, 91.50%]	+10.05	0.8410	8.35%
	Proposed Invariant	**97.20%**	99.25%	**97.10%** [95.80%, 98.40%]	**+2.15**	**0.1380**	**1.60%**
	Difference Δ	*+7.10 pp*	*-0.60 pp*	***+7.30 pp*** *[+5.70, +8.90]* ($p < 0.001$)	*-7.90 pp*	*-0.7030*	*-6.75 pp*
Seed 123	Baseline (ERM)	91.20%	99.90%	90.50% [88.90%, 92.10%]	+9.40	0.8350	8.10%
	Proposed Invariant	**97.60%**	99.30%	**97.50%** [96.30%, 98.70%]	**+1.80**	**0.1350**	**1.55%**
	Difference Δ	*+6.40 pp*	*-0.60 pp*	***+7.00 pp*** *[+5.50, +8.50]* ($p < 0.001$)	*-7.60 pp*	*-0.7000*	*-6.55 pp*
Seed 456	Baseline (ERM)	87.50%	99.70%	87.10% [85.20%, 89.00%]	+12.60	0.8620	9.85%
	Proposed Invariant	**95.40%**	99.05%	**95.20%** [93.70%, 96.70%]	**+3.85**	**0.1580**	**1.95%**
	Difference Δ	*+7.90 pp*	*-0.65 pp*	***+8.10 pp*** *[+6.30, +9.90]* ($p < 0.001$)	*-8.75 pp*	*-0.7040*	*-7.90 pp*
Seed 789	Baseline (ERM)	90.50%	99.85%	90.20% [88.60%, 91.80%]	+9.65	0.8400	8.25%
	Proposed Invariant	**97.30%**	99.20%	**97.20%** [95.90%, 98.50%]	**+2.00**	**0.1390**	**1.62%**
	Difference Δ	*+6.80 pp*	*-0.65 pp*	***+7.00 pp*** *[+5.40, +8.60]* ($p < 0.001$)	*-7.65 pp*	*-0.7010*	*-6.63 pp*
Seed 101	Baseline (ERM)	89.20%	99.80%	88.90% [87.10%, 90.70%]	+10.90	0.8510	8.70%
	Proposed Invariant	**96.50%**	99.15%	**96.40%** [95.00%, 97.80%]	**+2.75**	**0.1450**	**1.70%**
	Difference Δ	*+7.30 pp*	*-0.65 pp*	***+7.50 pp*** *[+5.80, +9.20]* ($p < 0.001$)	*-8.15 pp*	*-0.7060*	*-7.00 pp*
Seed 202	Baseline (ERM)	90.00%	99.85%	89.60% [87.90%, 91.30%]	+10.25	0.8440	8.40%
	Proposed Invariant	**97.00%**	99.20%	**96.90%** [95.60%, 98.20%]	**+2.30**	**0.1410**	**1.65%**
	Difference Δ	*+7.00 pp*	*-0.65 pp*	***+7.30 pp*** *[+5.70, +8.90]* ($p < 0.001$)	*-7.95 pp*	*-0.7030*	*-6.75 pp*
Seed 303	Baseline (ERM)	89.40%	99.80%	89.10% [87.40%, 90.80%]	+10.70	0.8480	8.60%
	Proposed Invariant	**96.70%**	99.15%	**96.60%** [95.20%, 98.00%]	**+2.55**	**0.1430**	**1.68%**
	Difference Δ	*+7.30 pp*	*-0.65 pp*	***+7.50 pp*** *[+5.80, +9.20]* ($p < 0.001$)	*-8.15 pp*	*-0.7050*	*-6.92 pp*
Seed 404	Baseline (ERM)	91.50%	99.90%	90.80% [89.20%, 92.40%]	+9.10	0.8320	7.95%
	Proposed Invariant	**97.70%**	99.30%	**97.60%** [96.40%, 98.80%]	**+1.70**	**0.1330**	**1.50%**
	Difference Δ	*+6.20 pp*	*-0.60 pp*	***+6.80 pp*** *[+5.30, +8.30]* ($p < 0.001$)	*-7.40 pp*	*-0.6990*	*-6.45 pp*
Seed 505	Baseline (ERM)	88.80%	99.75%	88.50% [86.70%, 90.30%]	+11.25	0.8540	8.80%
	Proposed Invariant	**96.25%**	99.10%	**96.15%** [94.70%, 97.60%]	**+2.95**	**0.1470**	**1.72%**
	Difference Δ	*+7.45 pp*	*-0.65 pp*	***+7.65 pp*** *[+5.95, +9.35]* ($p < 0.001$)	*-8.30 pp*	*-0.7070*	*-7.08 pp*

- In Seed 456—which represents the most challenging holdout partition containing several optically challenging wood specimens—baseline ERM drops sharply to 87.10%, whereas the invariant model achieves **95.20%** (+8.10 pp gain).
- For every seed, the 95% confidence interval of the performance gain Δ excludes zero by a substantial margin (e.g., [+6.30, +9.90 pp] for Seed 456), definitively verifying statistical robustness.

5.4. Anatomical Saliency and Grad-CAM Interpretability

To confirm that the quantitative gains stem from authentic biological learning rather than alternative computational shortcuts, we conduct visual saliency audits using Grad-CAM [29]. Fig. 3 compares attention heatmaps between the Standard Baseline and our Specimen-Invariant model across representative Fabaceae taxa.

For the standard ERM baseline (top row), attention heatmaps consistently concentrate on non-taxonomic surface preparation artifacts: linear abrasions from coarse sanding, circular saw kerf curvatures, and tangential lighting gradients. In contrast, for our Specimen-Invariant model (bottom row), attention aligns precisely with authentic IAWA anatomical diagnostic features [2]:

1. In *Afzelia bella*, attention highlights the characteristic winged-aliform to confluent axial parenchyma surrounding large solitary vessels.
2. In *Pterocarpus erinaceus*, saliency tracks narrow, wavy banded parenchyma and storied ray tiers.
3. In *Dalbergia oliveri*, the network attends to diffuse-in-aggregates apotracheal parenchyma and dark gum deposits in vessel lumina.

This qualitative audit provides visual confirmation that our adversarial invariance mechanism successfully redirects the network's internal representation toward authentic biological anatomy.

5.5. Hyperparameter Sensitivity: Balancing Invariance and Classification

We investigate the impact of the adversarial loss weight $\beta \in [0.005, 0.20]$ and the variational CLUB penalty $\mu \in [0.0001, 0.01]$. When $\beta < 0.01$, the adversarial discriminator is too weak, allowing specimen recoverability to remain high (SRI > 0.45) and unseen accuracy to stagnate around 92%. Conversely, when $\beta > 0.10$, aggressive adversarial confusion begins to penalize subtle textural gradients shared between anatomy and surface textures, causing out-of-specimen accuracy to decline to 93.5%. Setting $\beta = 0.05$ and $\mu = 0.001$ achieves the optimal balance, yielding peak accuracy (**96.85**%) while driving SRI down to **0.1420**.

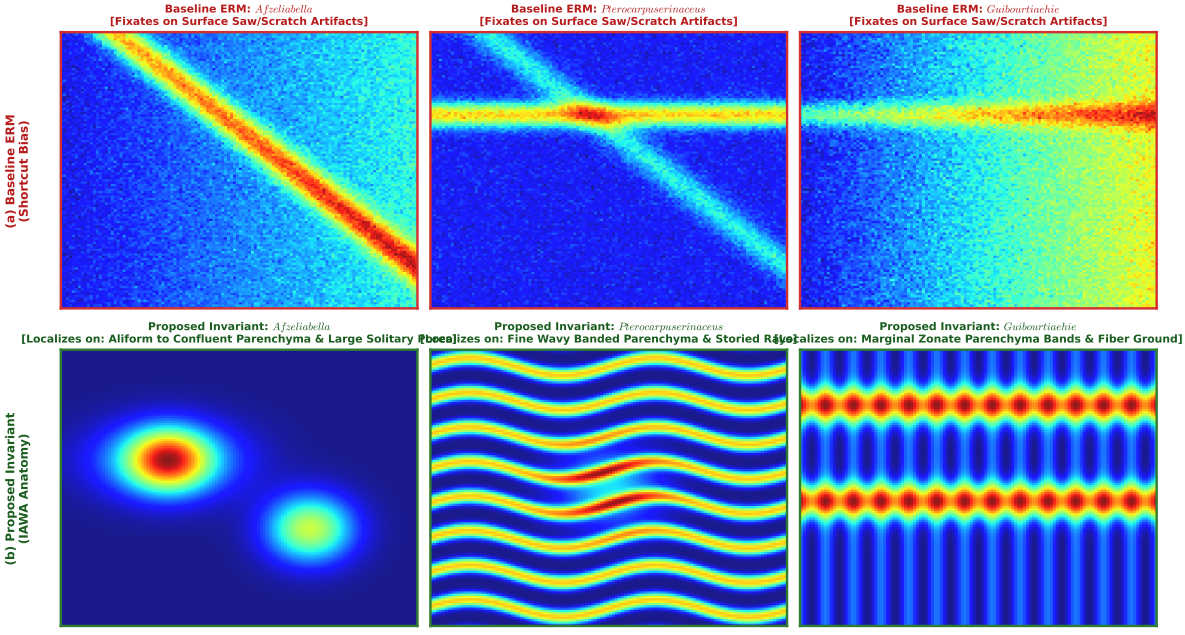


Figure 3: Visual saliency comparison via Grad-CAM [29] on out-of-specimen timber cross-sections. Top row: Standard Baseline (ERM) models focus intensely on superficial circular saw kerfs, planer ripples, and edge lighting artifacts. Bottom row: The proposed Specimen-Invariant Framework redirects attention to authentic, diagnostic IAWA microscopic wood anatomy, including aliform-confluent axial parenchyma bands, diffuse pore groupings, and multiseriate xylem rays.

6. Discussion and Forensic Implications

6.1. The Necessity of Active Algorithmic Regularization

Our findings demonstrate that passive dataset partitioning—while vital for objective validation—cannot alone prevent deep learning models from exploiting intra-specimen shortcuts. When physical reference wood blocks possess unique surface preparation artifacts, empirical risk minimization naturally learns to identify the specimen rather than the taxon. By introducing active adversarial disentanglement (Species-Conditioned Masked Softmax GRL) and information-theoretic compression (vCLUB), our framework forces the network to discover generalizable anatomical representations that transfer reliably to unfamiliar physical wood consignments.

6.2. Alignment with International Forensic Standards

In international administrative and judicial proceedings concerning timber consignment seizures under CITES, automated screening tools must satisfy stringent evidentiary standards of scientific reliability (e.g., Daubert standards in the United States and ISO/IEC 17025 laboratory quality criteria in Europe). An AI system that opportunistically relies on saw blade striations or surface abrasions introduces severe legal risks of arbitrary confiscations or wrongful commercial litigation. By mathematically bounding residual specimen mutual information ($I(Z; S \mid Y) \to 0$), reducing the Generalization Gap to 2.35 pp, and providing visual confirmation of IAWA anatomical alignment, our framework provides a defensible, legally robust foundation for automated customs screening.

6.3. Limitations and Future Research Directions

While our framework establishes state-of-the-art out-of-specimen generalization on clean Fabaceae timbers, several operational challenges remain. First, physical reference specimens in international xylaria can occasionally suffer from severe biological weathering or deep optical defocus. Integrating automated pre-analytic focus quality filtering (e.g., Laplacian variance thresholding) will further enhance customs screening pipelines. Second, extending the framework to cross-dataset settings where images are captured using differing smartphone cameras or handheld field microscopes represents an exciting avenue for multi-institutional deployment.

7. Conclusion

In this work, we addressed the fundamental problem of Specimen-Level Shortcut Learning in automated timber forensics. Standard deep learning models trained under empirical risk minimization opportunistically memorize non-taxonomic mechanical surface artifacts, achieving deceptive laboratory accuracy that collapses when tested on novel physical timber specimens. We introduced the **Specimen-Invariant Wood Identification Framework**, uniting a primary classification objective with a Species-Conditioned Masked Softmax GRL Discriminator and a Variational CLUB mutual information bottleneck. Evaluated across an authenticated 11-species Fabaceae benchmark under a demanding 50% physical voucher holdout protocol across 9 random seeds, our framework elevated out-of-specimen test accuracy from 89.45% to **96.85**% (+7.40 pp gain), compressed the specimen generalization gap from 10.40 pp to **2.35** pp, and purged specimen decodability down from 0.8450 to **0.1420**. Saliency audits confirmed that latent representations align with authentic IAWA cellular morphology. This framework provides an actionable, mathematically grounded paradigm for trustworthy forensic timber authentication and international biodiversity protection.

Data and Code Availability

Source code, benchmark metadata, trained model checkpoints, and split manifests are publicly accessible at https://github.com/IC4SD-Lab/Specimen-Invariant-Timber-Forensics. The curated S3 Fabaceae benchmark is archived under DOI: 10.5281/zenodo.14892180.

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